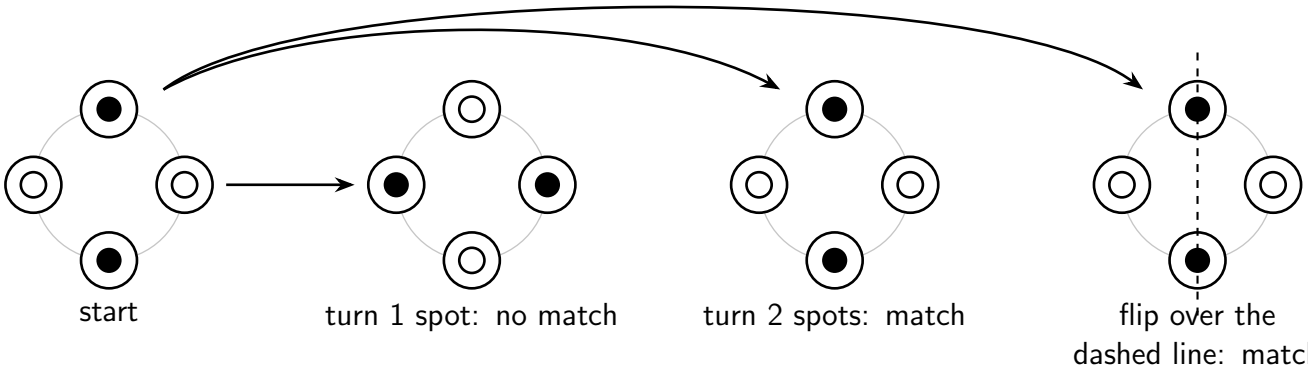
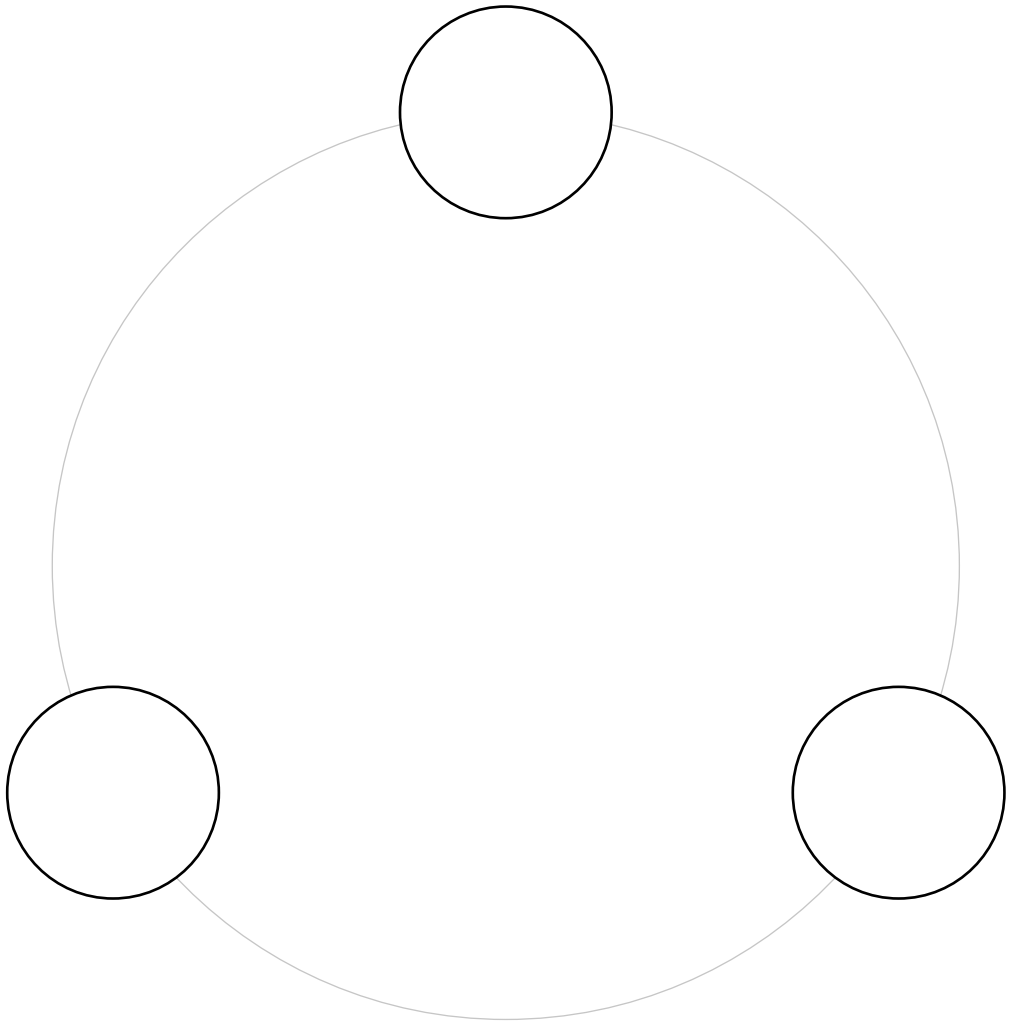
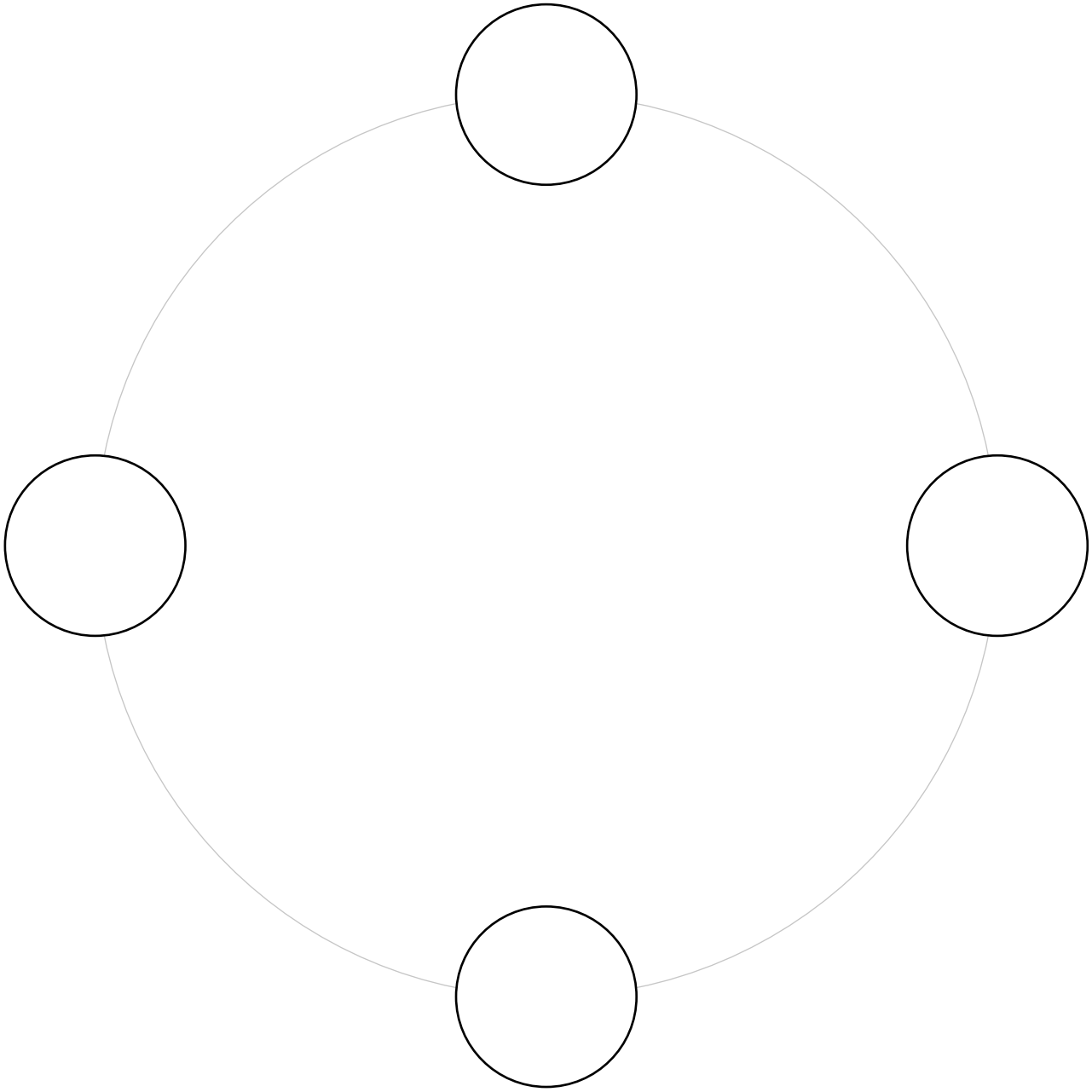
Fill every spot; neighbors may be the same kind. Only counter kinds and spots count. Keep the original still. Turn or flip (turn over) its tracing copy until all spots line up. A whole turn does not count.



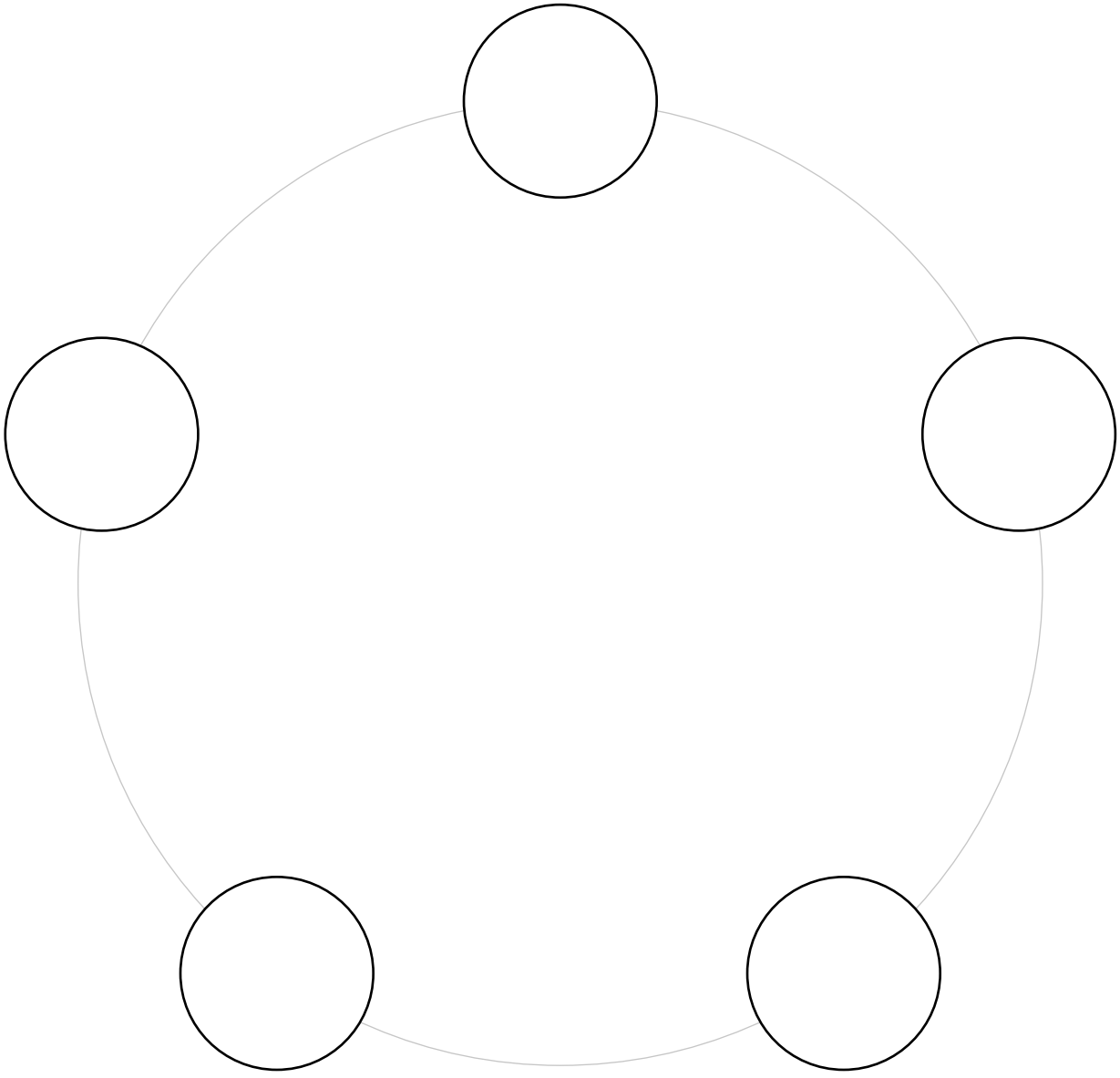
Problem 1: Find how few kinds of counter can make a pattern on this ring that no turn or flip can match.



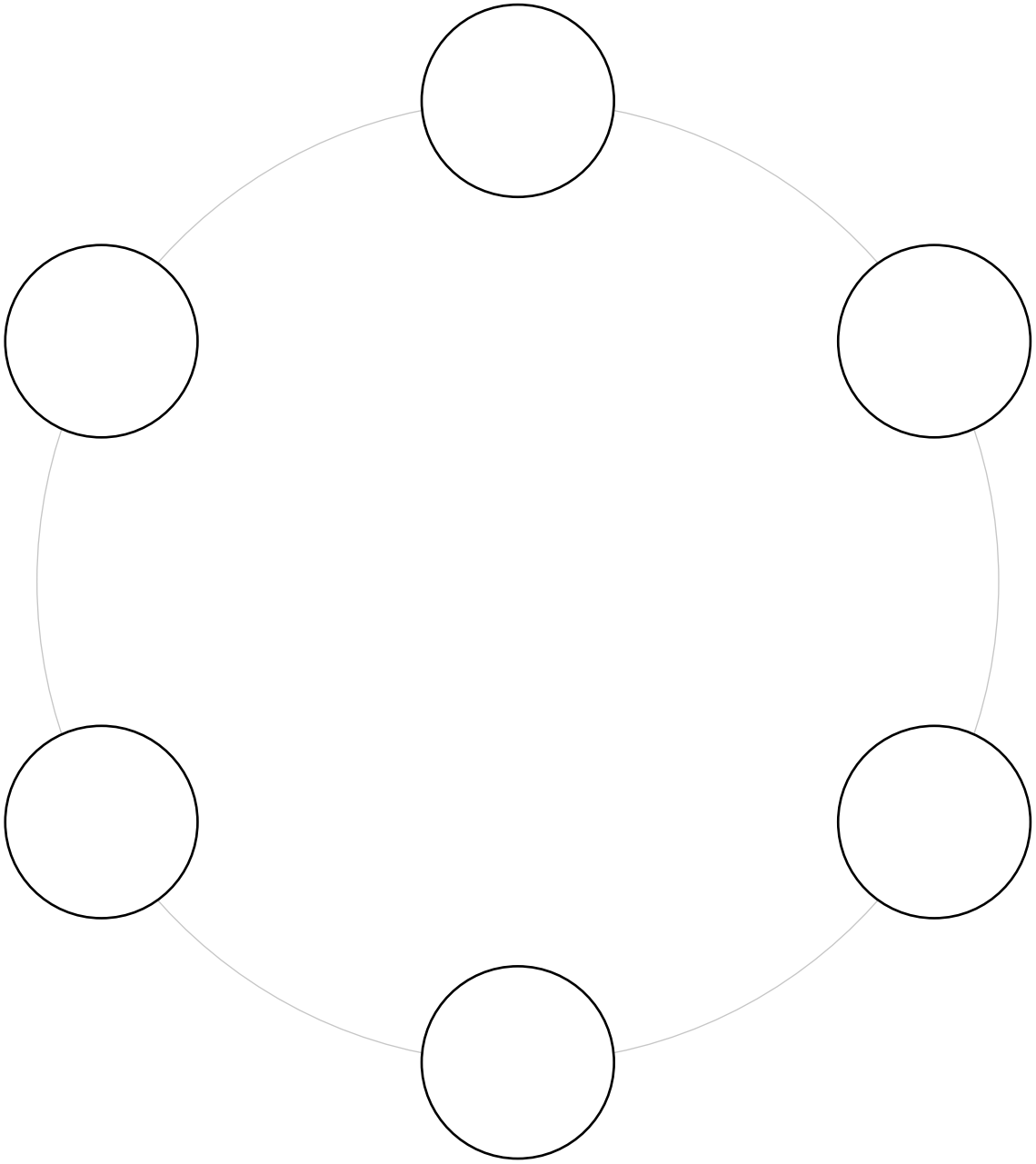
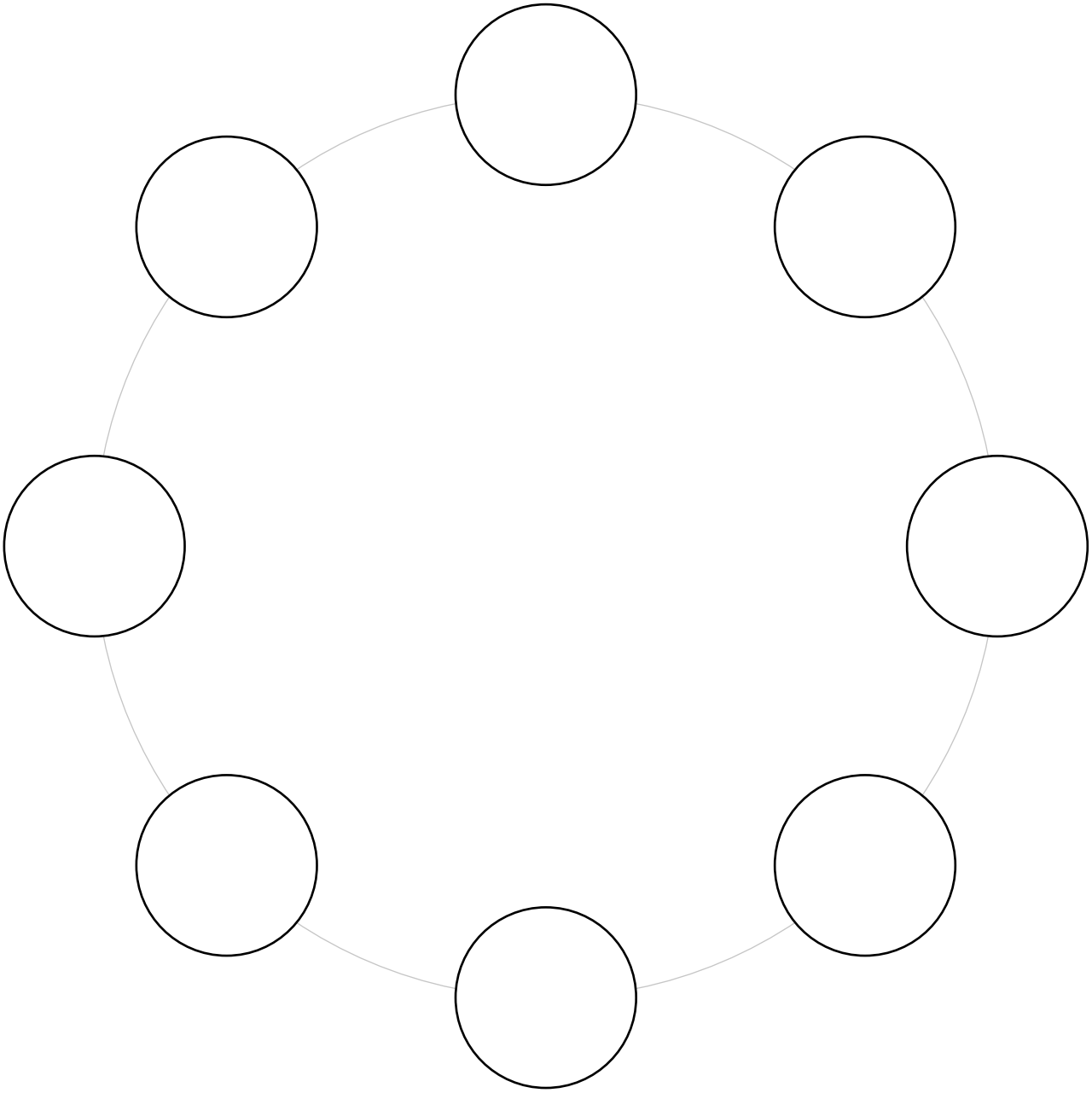
Problem 2: Find how few kinds of counter can make a pattern on this ring that no turn or flip can match. Let someone test your best pattern.



Problem 3: Find how few kinds of counter can make a pattern on this ring that no turn or flip can match. Can you explain why fewer kinds would fail?



Problem 4: Find a six-ring pattern using only two kinds of counter that no turn or flip can match. Show how you know every possible move fails.



Problem 5: Find an eight-ring pattern with two kinds of counter that no turn or flip can match. How few counters of one kind can it have?